FULL CARD 23 — Reaction Rates and Collision Theory

CHEMISTRY • BALANCED EQUATIONS • DATA • EXPERIMENT DESIGN

CORE MODEL • A chemical reaction rearranges atoms. In a closed system, atoms are conserved; total mass remains constant when all reactants and products are included.

Reaction: calcium carbonate + hydrochloric acid → calcium chloride + water + carbon dioxide

$$CaCO_3 + 2HCl \rightarrow CaCl_2 + H_2O + CO_2$$

Experiment: effect of acid concentration on reaction rate

Trial	HCl concentration (mol/L)	$CaCO_3$ mass (g)	Time to collect 40 mL CO_2 (s)	Rate proxy 40 ÷ time (mL/s)
A	0.5	2.0	80	0.50
B	1.0	2.0	43	0.93
C	1.5	2.0	29	1.38
D	2.0	2.0	22	1.82

Interpretation

Higher concentration usually means more reacting particles per unit volume, increasing collision frequency. More frequent successful collisions can increase reaction rate. The table supports a trend, but it does not establish a precise rate law.

Variables and quality

Independent variable: acid concentration. Dependent variable: time to collect 40 mL of gas. Controls: mass and surface area of calcium carbonate, acid volume, temperature, apparatus, and endpoint definition. Repeat each trial and calculate a mean; check for gas leaks.

Worked example: Trial C rate proxy = 40 mL ÷ 29 s ≈ 1.38 mL/s. This is an average over the measured interval, not necessarily the initial rate.

Safety and precision

Wear eye protection; hydrochloric acid is corrosive. Use a stable apparatus and appropriate quantities under supervision. A gas syringe can improve measurement compared with estimating bubbles. Do not seal a reaction vessel unless the apparatus is designed for pressure.

Practice: Predict what could happen if powdered calcium carbonate replaces chips. Explain the prediction using collision theory and name one additional control variable.